

Nils Matteson

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Data Science & CS graduate (UW–Madison, 2026) and incoming CS master’s student (Northeastern, Silicon Valley). Founder of **thaw** (LLM-inference infrastructure) and **Matteson Systems LLC**, with experience across GPU/CUDA inference, distributed systems, and applied ML.

EDUCATION

Northeastern University <i>M.S. Computer Science, Silicon Valley campus</i>	Sep 2026 – May 2028 San Jose, CA
University of Wisconsin–Madison <i>B.S. Data Science, Minor in Computer Science</i>	May 2026 Madison, WI
– Coursework: Big Data Systems, Artificial Intelligence, Machine Learning, Machine Organization, Programming III (Data Structures), Linear Algebra, Discrete Math, Data Science Modeling.	

TECHNICAL SKILLS

Languages: Python, Rust, Go, C++, CUDA, TypeScript/JavaScript, SQL, R, Java
ML & Inference: PyTorch, vLLM, SGLang, Triton, Hugging Face, XGBoost, scikit-learn, RAG, conformal prediction, NumPy/Pandas
Systems & Infra: CUDA/PyO3 FFI, gRPC, Raft, LSM-trees, Kafka, AWS (Bedrock/S3), Docker, PostgreSQL, Redis, CI/CD, Linux
Web & Data: Next.js, React, Node, WebSockets, DeckGL/MapLibre, Drizzle, Flask, DuckDB

EXPERIENCE

thaw (startup) <i>Founder & Lead Engineer</i>	2026 – Present Rust · CUDA · Python
– Architected thaw , a Rust + CUDA library that snapshots and restores live LLM inference state (weights, KV cache, prefix-hash table, scheduler), enabling sub-second forking of vLLM sessions: 0.88s median fork vs. ~340s cold boot on H100 (committed benchmark receipts). – Built a double-buffered 0_DIRECT DMA pipeline overlapping disk reads, PCIe transfer, and an 8-shard parallel CRC32C verifier that matches a serial pass exactly, hitting 14.3 GB/s weight restore and 3.4× faster 70B load with bit-identical output across 8 models . – Eliminated a 60× KV-cache snapshot bottleneck by coalescing ~16K tiny per-block DMAs into one contiguous gather, then reconstructing vLLM’s prefix-cache hash table on restore so cold-started requests skip prefill. – Designed a CudaBackend trait (mock + real CUDA behind one contract) running the full pipeline and 172 Rust tests on macOS with no CUDA toolchain; opened vLLM RFC #34303 (co-founded with M. Yu, K. Kapur).	
Matteson Systems LLC <i>Founder & Engineer</i>	2026 – Present Next.js · Postgres · Playwright · Claude
– Built and shipped an autonomous SMB cold-outreach product that scraped and scored 10,500+ local businesses (OpenStreetMap) and surfaced 158 high-priority leads in a single run. – Engineered a two-stage Claude-vision audit pipeline combining real-device Playwright screenshots (Chromium + WebKit/iPhone), live Lighthouse Core Web Vitals, and on-page analysis into per-lead website critiques and personalized cold emails at ~\$0.04/lead . – Designed an append-only event-sourced Postgres CRM (6 tables, 8 Drizzle migrations) and a threaded Gmail drip on Vercel cron (RFC822 in-thread replies, reputation-aware send caps) with a per-call cost ledger in integer microcents.	
Research Cyberinfrastructure, UW–Madison DoIT <i>AI Workflows Research Assistant</i>	Jan 2026 – Apr 2026 AWS Bedrock · RAG · LLM eval
– Built the evaluation and cost-tracking framework benchmarking 9 LLMs and 3 ensemble strategies on AWS Bedrock across 282 questions; ensemble majority voting (0.840) outperformed every individual model. – Showed via Pareto analysis that Llama-4 Maverick reaches 98% of top accuracy at a fraction of cost and latency, informing production model selection; presented at the UW–Madison ML+X Forum.	

SELECTED PROJECTS

Sentinel: Distributed Message Queue github.com/matteso1/sentinel	Go
– Built a Kafka-inspired distributed log engine from scratch: custom LSM-tree storage (skip-list memtables at 3.9M reads/s, SSTables w/ CRC32, WAL, leveled compaction), a gRPC wire protocol, and topic/partition consumer groups. – Implemented Raft consensus from scratch (leader election, log replication, split-brain prevention), validated by a deterministic in-memory network simulator covering partition + heal; 45 passing tests.	
Madison Metro ML: Real-Time Bus Arrival Prediction madisonbuseta.com	Python, XGBoost, React
– Shipped a live ML system correcting transit-API ETAs with a 47-feature XGBoost model and Mondrian conformal prediction (calibrated 90%-coverage intervals stratified by route × day-type × horizon). – Automated nightly retraining (GitHub Actions) behind a hard deployment gate (≥2s MAE gain, no metric regression); React/DeckGL/MapLibre map rendering 200+ live vehicles at 60fps.	
Also: Lattice (Rust+PyO3 reactive framework, Canelift JIT, 10.6× vs. Streamlit) · brain2text (BCI Kaggle: 5-layer GRU + CTC decoder, 40% WER) · gitstare (Rust TUI, on crates.io) · LockBox (AES-256-GCM password manager). Coursework projects: distributed-systems visualizations (Cassandra/Kafka/HDFS/HBase, CS544) · 1.2M-tweet sentiment pipeline on the UW HTCondor grid (STAT405) · causal-inference propensity-score matching in R (STAT479).	